

Nikiforos Mandilaras

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Work Experience

Taboola

Machine Learning Engineer - Tech Lead

Remote - Subcontractor at Deeplab

Nov. 2021 – Present

- Working with state of the art algorithms and technologies in a large scale Recommender System.
- Applying LLMs for representation learning, leveraging fine-tuned embeddings in multiple downstream tasks.
- Developed a new candidate generation model using Vespa, leveraging a two-tower architecture to efficiently provide online ranking of suitable items.
- Delivered an incremental learning framework with 12x faster training and deployment cycles, achieving revenue lift.
- Part of the algo expert team, leading high-priority initiatives, designing workflows, and providing guidance and code reviews for the rest of algo department.

Ubitech

Machine Learning Engineer

Remote

June 2021 – Nov. 2021

- Participated in research activities and designed solutions for improving efficiency in smart-grid environments, including self-consumption optimization, energy savings estimation, and flexibility prediction.

Intracom Telecom

Data Scientist / Big Data Engineer

Athens, Greece

May 2018 – May 2020

- Designed and implemented machine learning algorithms, concerning new smart features for company's products (e.g. load forecasting, grid optimisation, resource allocation).
- Developed Big Data applications for telco and banking clients using the Hadoop ecosystem, including high-throughput ingestion pipelines and complex data analysis.

Education

National Technical University of Athens

Athens, Greece

o MSc. in Data Science and Machine Learning, 9.61/10

Oct. 2018 – Aug. 2020

Master Thesis in Resource Allocation with Deep Reinforcement Learning Methods.

o Integrated Master in Electrical and Computer Engineering, 8.44/10

Sept. 2012 – Feb. 2018

Major in Computer Science: Software, Computer Systems & Networks - Master Thesis in the area of Big Data.

Technical Skills

Prog. Languages: Python, Java

DataBases: BigQuery, Vespa, MongoDB, Elastic

Deep Learning: PyTorch, TensorFlow, Keras, LLMs

CI/CD: Airflow, Docker, Jenkins, GitLab, Graphana

ML & Big Data: Apache Spark, Kafka, HBase, Sklearn, Pandas, Numpy

Practises: Design Patterns, Agile Methodologies

App Frameworks: Spring, Django

Selected Projects

Deep Reinforcement Learning: Developed a RL agent for managing last-level cache sharing in multi-core systems, improving SLA performance on a critical service and accelerating best-effort applications by 4x.

Self Supervision: Created sophisticated embeddings that improved recommendation accuracy, boosting revenue for users with poor signal history.

LLM-Based Embedding Pipeline: Designed and built a distributed production pipeline that generates item embeddings for ad recommendations. Used LLMs for description enrichment and embedding generation. Powers downstream recommendation models with continuously updated item representations.

Additional Info

Languages: English (C2) & French (B2)

Honors: C. Papakyriakopoulos Math Award – N. Kritikos Scholarship

Military Service: Fulfilled Sept. 2020 - June 2021

Publications

P. Katsileros, N. Mandilaras et. al. An Incremental Learning framework for Large-scale CTR Prediction. 16th ACM Conference on Recommender Systems. RecSys '22 - Seattle, WA, USA.